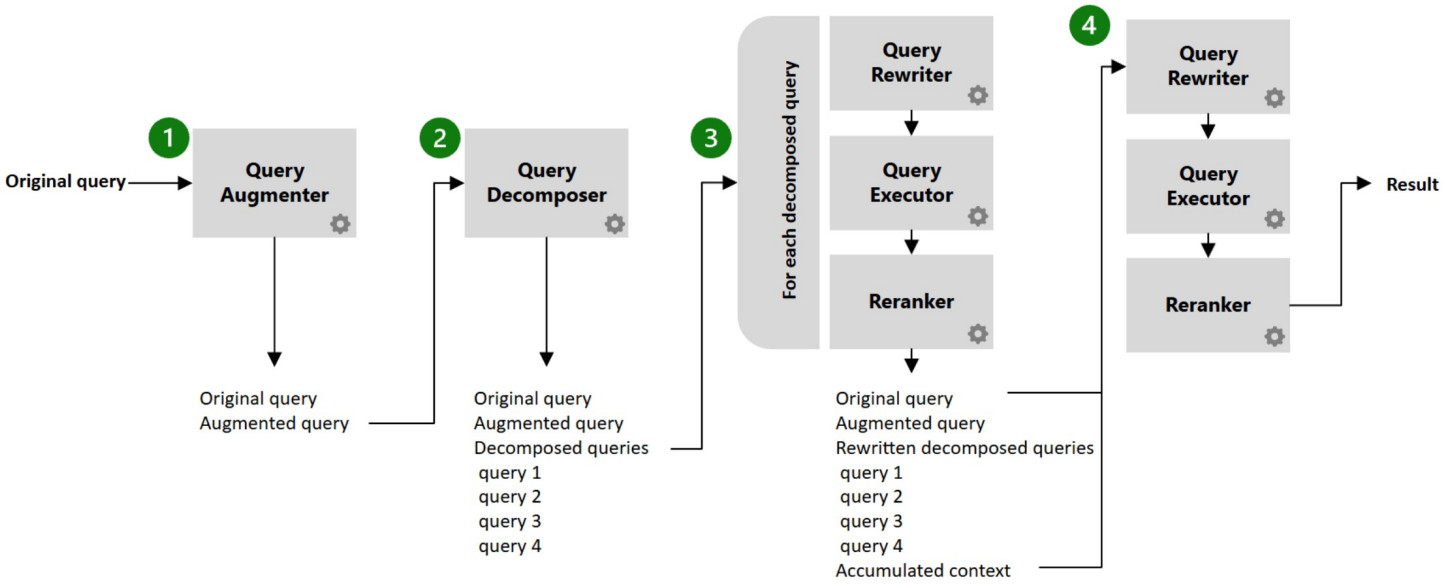


Best Batch

Ai adoption

2025 - 2026

ML ✓

↳ DL → genAi ✓

↳ agentic Ai ✓

Traditional Route → LLM's

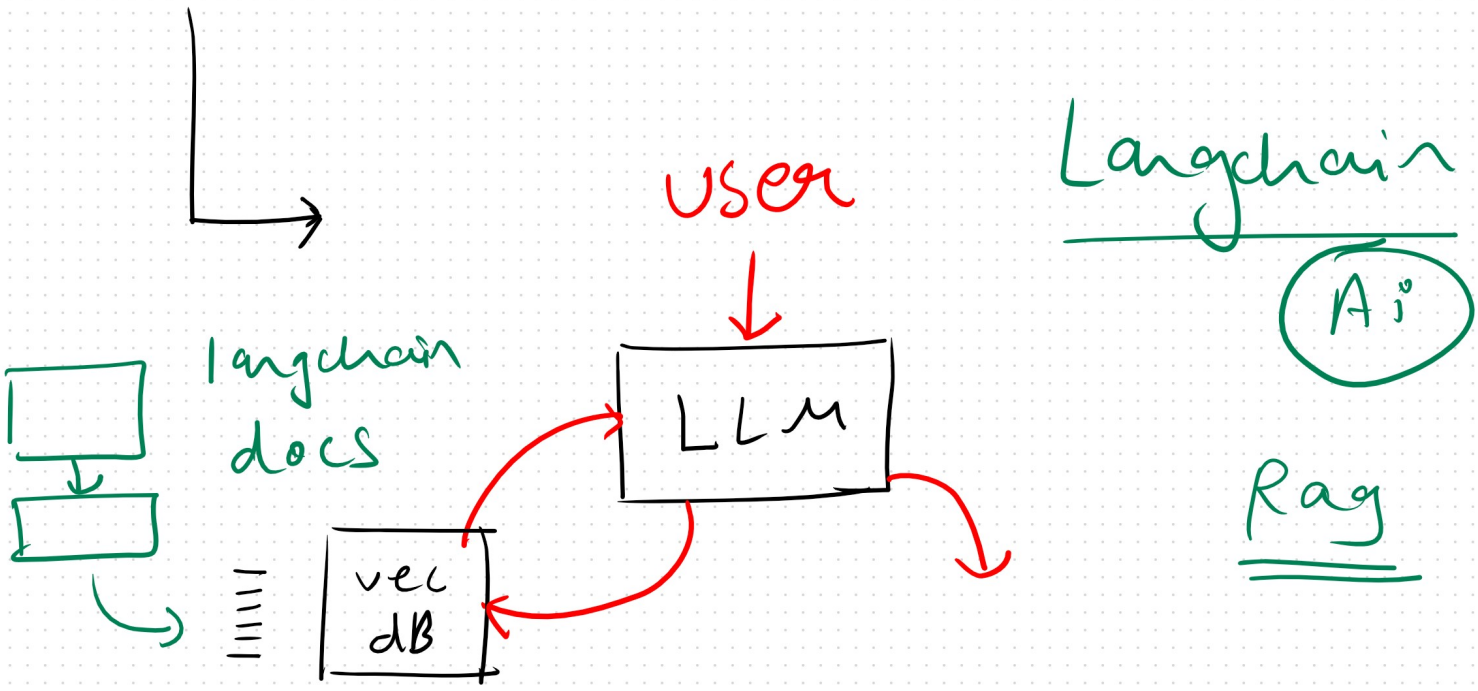
Modern Route → use LLM's

Advanced Route → make LLM's

Catch up →

Rag

Ai agent



Langchain

Ai

Rag

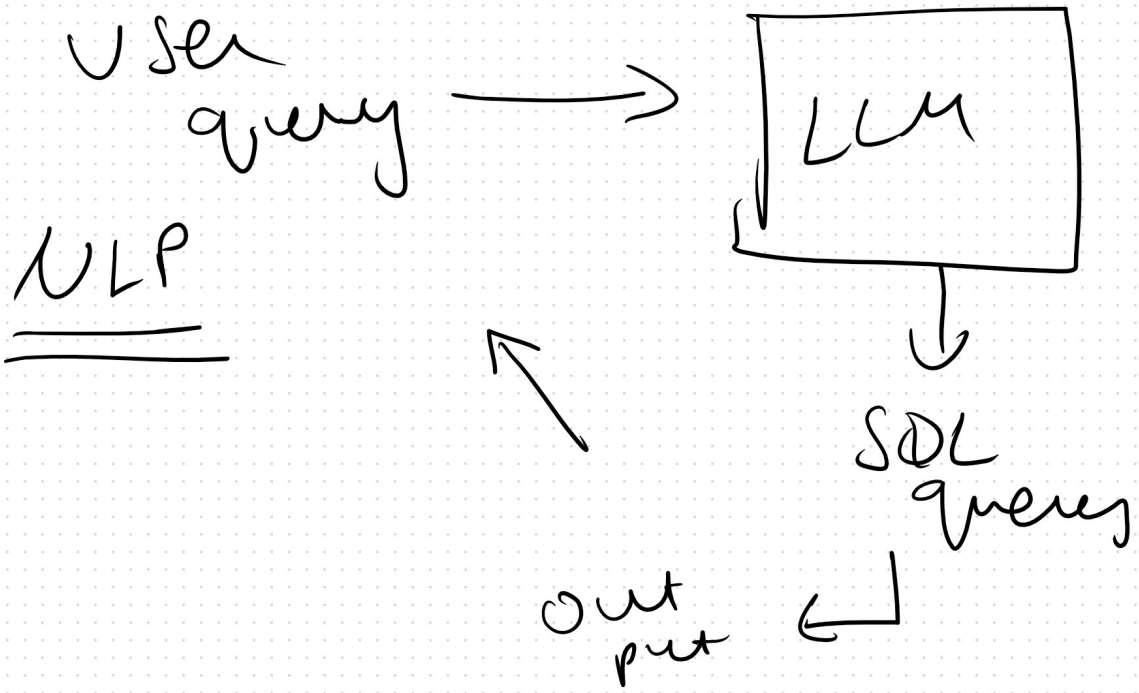
Text2 SQL

Select * from amazon
where page = "fashion"

Select * for men clothes

where jean = Baggy &

price ≤ 2000;



1 min

1 sec

Ai agent

24 + 36

digital human.

LLM

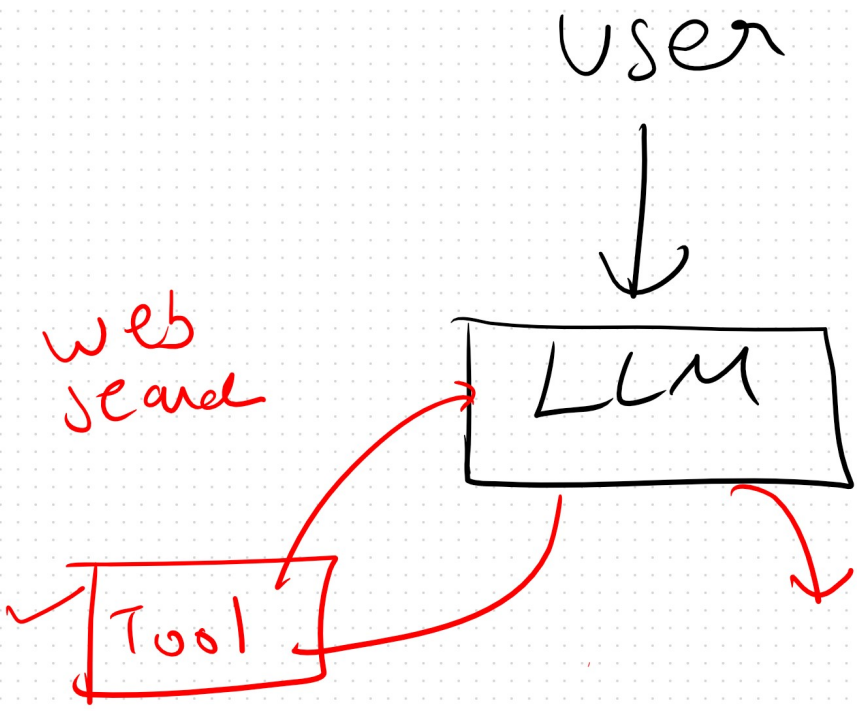
memory

py function

Brain

Memory

Tools



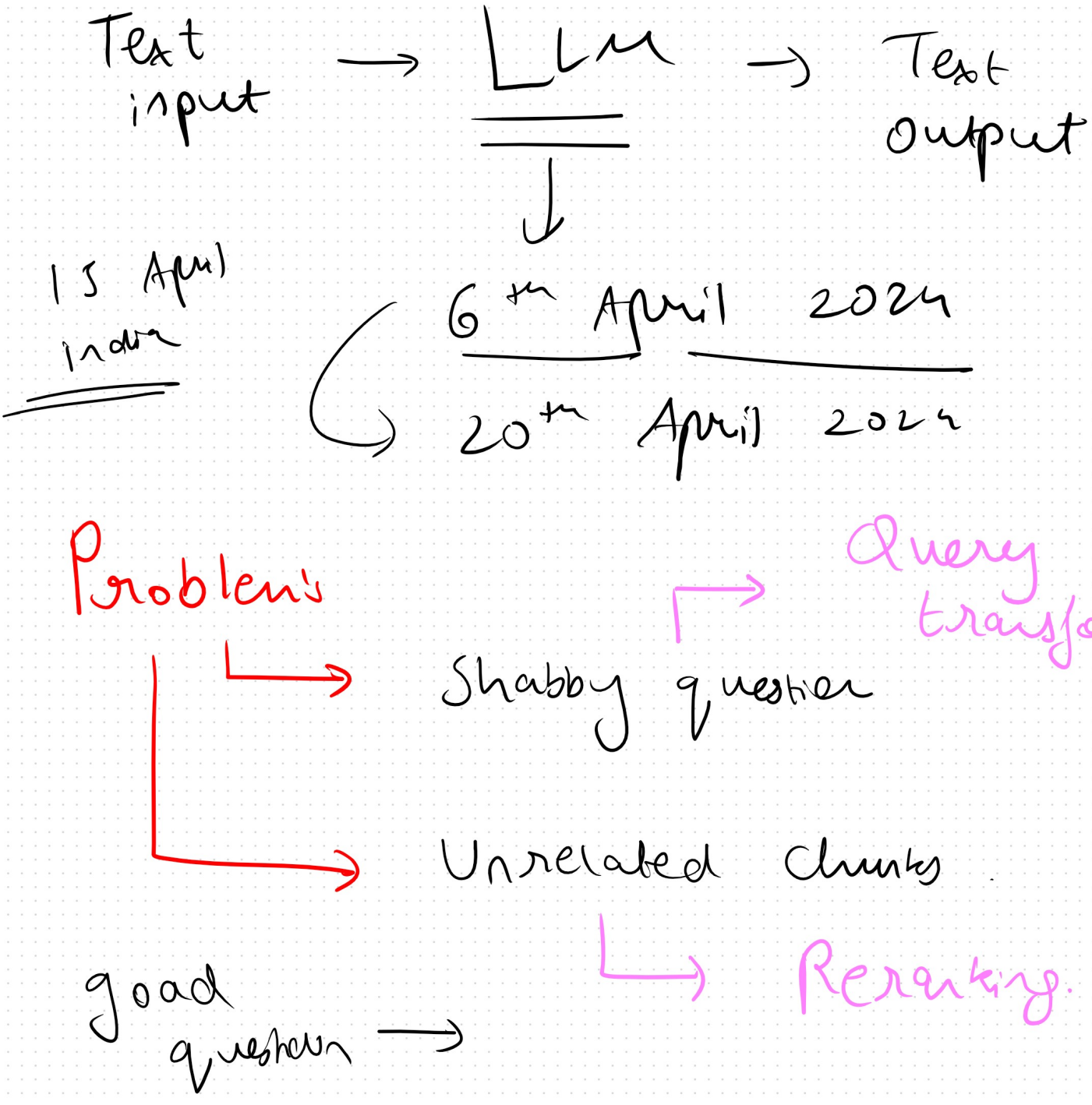
When is diwali in india in 2026

7 - 8

Agenda ....

User → LLM → rewrite

1. Query transformation
2. Observability → LLM call → Tool → token cost
3. MCP

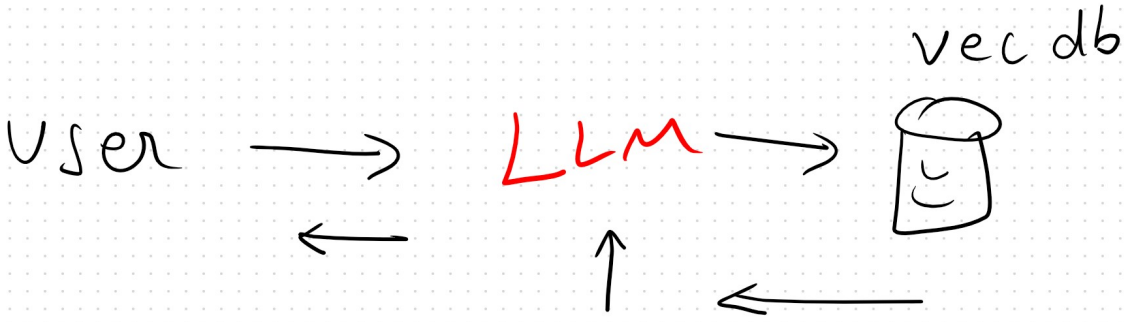

Test input
LLM
Test output
15 April India
6th April 2024
20th April 2024
Problems
Query transformation
Shabby question
Unrelated Chunks
Reranking.
good question

.env file

Brain

Text → Text

openai api key



Openai → chatgpt

antropic → claude

google → gemini

google cloud → vertex ai

aws → Bedrock

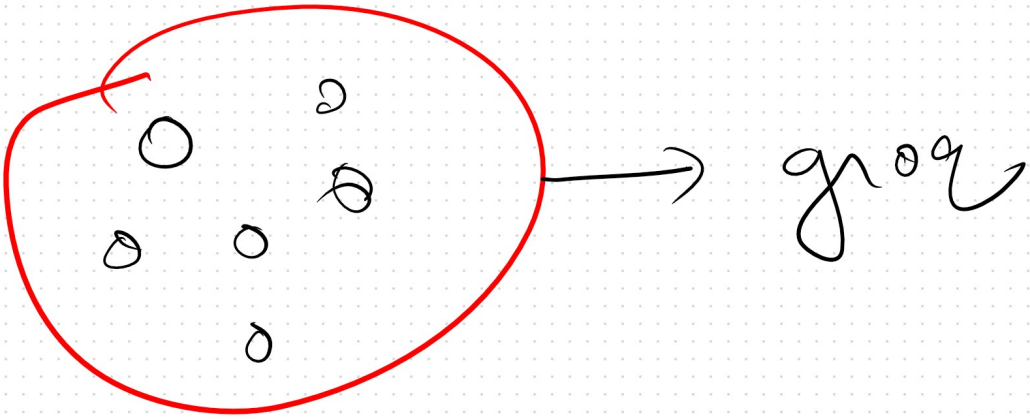
azure → foundry.

Open source

- deepseek
- mixtral
- qwen.

LLM

- web
  - market
  - aggregator
    - groq,
    - openrouter
- local
  - ollama



Tracing

Chat.openai.com

LT → True

LE → app → langsmith

LAPI → auth

✓ Langsmith → observability !

↳ LLM executes

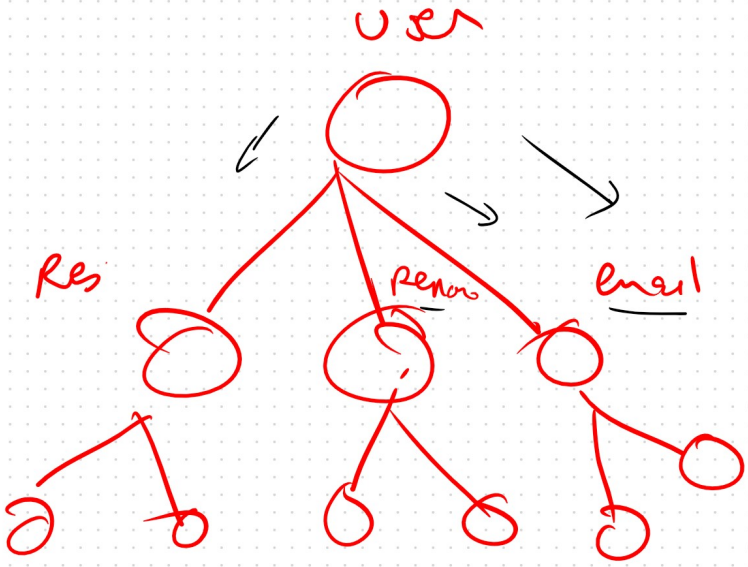
- each LLM call ↳ Trace.

Basics :-

Basic :-

1. Span → each unit of trace
2. Trace
3. Waterfall → execution time

- input • 1s
- LLM • 3s
- Prompt 0.1
- VectorDB 2
- LLM
- Response 0.3



what is ✓ neural network ✓

available docs

Query routing



RL

Intro NN

ML

NLP

Deep learning Overly.

① Multi query generation ~

② Query routing ~

③ Query compression ~

web , vectdb , doc

Assignment

↳ notebook !

30 mins

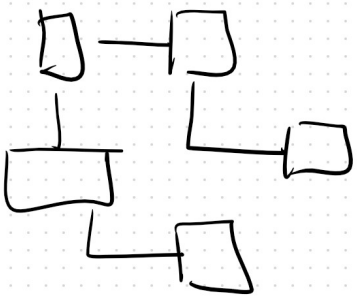
Ragas !

Rag evaluator

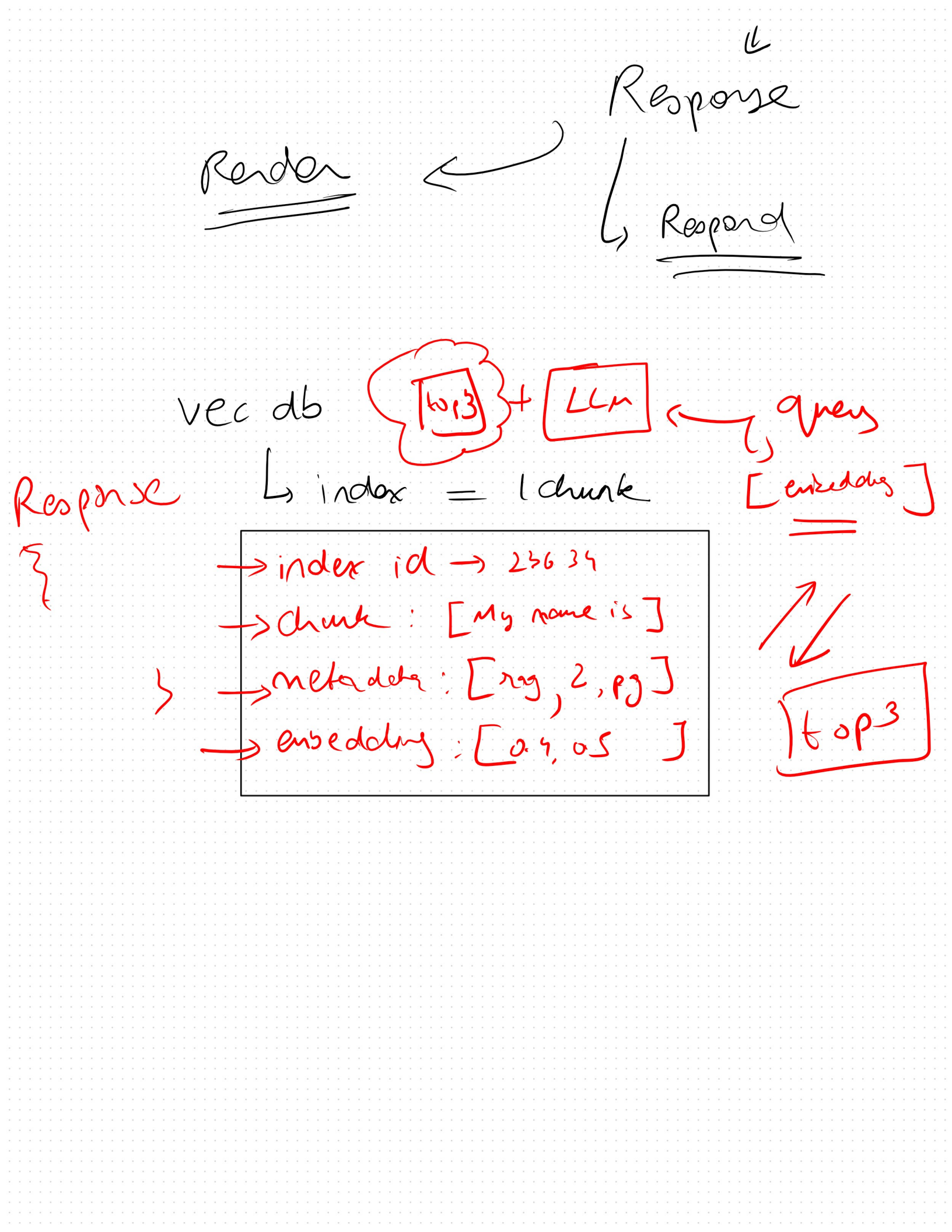
SQL

user → LLM

↳



↳ Mysql


Response
Render
Respond
vec db
top3
+
LLm
query
[embedding]
Response
↳ index = 1 chunk
→ index id → 2364
→ chunk : [My name is]
→ metadata : [rag, 2, pg]
→ embedding : [0.4, 0.5]
top3